

5.2 Classwork: Calculus Review (no calculator) — Markscheme

Total: 38 marks. All four problems are past IB AA HL questions; the mark allocations follow the official markschemes.

1. [Maximum mark: 8]

AA HL 2021 May P2 TZ2 Q9

$$(a) (1+t)^{-1} = 1 - t + t^2 \quad \mathbf{A1}$$

Accept the three terms listed: 1, $-t$, t^2 .

$$(b) \sec x = \frac{1}{\cos x} = \frac{1}{1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots} \quad \mathbf{(M1)}$$

$$\text{Let } t = \cos x - 1 = -\frac{x^2}{2} + \frac{x^4}{24} - \dots, \text{ so } \sec x = 1 - t + t^2 \quad \mathbf{(M1)}$$

$$= 1 - \left(-\frac{x^2}{2} + \frac{x^4}{24}\right) + \left(-\frac{x^2}{2} + \frac{x^4}{24}\right)^2 + \dots \quad \mathbf{A1}$$

$$= 1 + \frac{x^2}{2} - \frac{x^4}{24} + \frac{x^4}{4} + \dots \quad \mathbf{A1}$$

$$= 1 + \frac{x^2}{2} + \frac{5x^4}{24} \quad \mathbf{AG}$$

The second A1 needs both x^4 contributions, $-\frac{x^4}{24}$ and $+\frac{x^4}{4}$ (from squaring $-\frac{x^2}{2}$), seen before the given answer. Condone the absence of “...”.

Differentiating $\sec x$ four times earns no marks: the question requires the method of parts (a) and (b).

$$(c) \arctan 2x = 2x - \frac{(2x)^3}{3} + \dots$$

$$\lim_{x \rightarrow 0} \frac{x \left(2x - \frac{8x^3}{3} + \dots\right)}{\left(1 + \frac{x^2}{2} + \frac{5x^4}{24}\right) - 1} \quad \mathbf{M1}$$

$$= \lim_{x \rightarrow 0} \frac{2x^2 - \frac{8x^4}{3} + \dots}{\frac{x^2}{2} + \frac{5x^4}{24}} = \lim_{x \rightarrow 0} \frac{2 - \frac{8x^2}{3} + \dots}{\frac{1}{2} + \frac{5x^2}{24}} \quad \mathbf{A1}$$

$$= 4 \quad \mathbf{A1}$$

Do not award the M1 unless x is replaced by $2x$ in the arctan series. Condone a missing “lim” and errors in the higher-order terms. Only the leading terms $2x^2$ and $\frac{x^2}{2}$ are needed for the answer.

L'Hôpital's rule gives the same limit but earns no marks: the question requires the Maclaurin series.

2. [Maximum mark: 16]

AA HL 2021 Nov P1 Q10

$$v(t) = 4 + 4t - 3t^2, \quad 0 \leq t \leq 3; \quad P \text{ at } O \text{ when } t = 0.$$

(a)(i) Valid approach to find the turning point: $v'(t) = 4 - 6t = 0$ (M1)

$$t = \frac{2}{3} \text{ (s)} \quad \text{A1}$$

Accept $t = -\frac{b}{2a} = -\frac{4}{2(-3)}$, or the average of the roots $\frac{-\frac{2}{3} + 2}{2}$.

(a)(ii) Attempt to integrate v (M1)

$$\int (4 + 4t - 3t^2) dt = 4t + 2t^2 - t^3 (+c) \quad \text{A1A1}$$

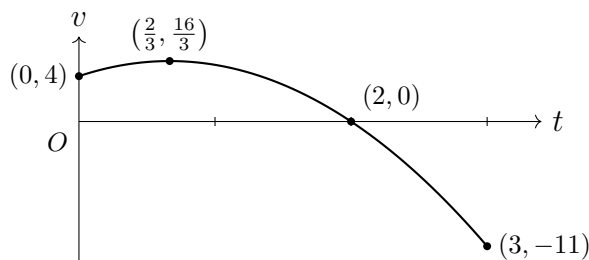
A1 for $4t + 2t^2$, A1 for $-t^3$.

Since $s(0) = 0$, $c = 0$. Substitute $t = \frac{2}{3}$: (M1)

$$4\left(\frac{2}{3}\right) + 2\left(\frac{2}{3}\right)^2 - \left(\frac{2}{3}\right)^3 = \frac{8}{3} + \frac{8}{9} - \frac{8}{27} \quad \text{A1}$$

$$= \frac{72 + 24 - 8}{27} = \frac{88}{27} \text{ (m)} \quad \text{AG}$$

The A1 needs the three fractions (or an equivalent line) before the given answer.

(b) Solve $4 + 4t - 3t^2 = 0$: $(2 - t)(2 + 3t) = 0$ (M1)Correct t -intercept at $t = 2$ A1Correct domain from 0 to 3, starting at $(0, 4)$ A1Vertex in approximately the correct place: $t = \frac{2}{3}$, $v > 4$ A1

The last two A marks are awarded only if the shape is a concave-down parabola; they are independent of each other and of the (M1). The end at $t = 3$ must be clearly shown; the value $v(3) = -11$ need not be labelled. The root $t = -\frac{2}{3}$ is outside the domain and must not be drawn.

(c) Integrate over $[0, 2]$ and $[2, 3]$ (P turns at $t = 2$) (M1)

$$\int_0^2 (4 + 4t - 3t^2) dt = 8 \quad \mathbf{A1}$$

$$\int_2^3 (4 + 4t - 3t^2) dt = 3 - 8 = -5 \quad \mathbf{A1}$$

Valid approach to sum the two distances: $\int_0^2 v dt - \int_2^3 v dt$ **(M1)**

Total distance travelled = $8 + 5 = 13$ (m) **A1**

$\int_0^3 v dt = 3$ is the final displacement, not the distance; it earns no marks on its own.

Accept $\int_0^3 |v| dt$ as the (M1) and the second (M1).

FT: award the A marks from the candidate's own antiderivative in (a)(ii) and root in (b).

3. [Maximum mark: 7]

AA HL 2021 May P1 TZ2 Q9

$$u = \sin x \Rightarrow du = \cos x \, dx \quad \text{A1}$$

$$\int \frac{\sin x \cos x}{\sin^2 x - \sin x - 2} dx = \int \frac{u}{u^2 - u - 2} du \quad \text{A1}$$

Attempt to use partial fractions: M1

$$\frac{u}{(u+1)(u-2)} \equiv \frac{A}{u+1} + \frac{B}{u-2} \Rightarrow u \equiv A(u-2) + B(u+1)$$

Valid attempt to solve for A and B (e.g. $u = 2$, $u = -1$) (M1)

$$A = \frac{1}{3} \text{ and } B = \frac{2}{3} \quad \text{A1}$$

$$\int \left(\frac{1}{3(u+1)} + \frac{2}{3(u-2)} \right) du = \frac{1}{3} \ln|u+1| + \frac{2}{3} \ln|u-2| (+C) \quad \text{A1}$$

$$= \frac{1}{3} \ln|\sin x + 1| + \frac{2}{3} \ln|\sin x - 2| + C \quad \text{A1}$$

Condone the absence of $+C$ or of the moduli in the u line, but not in the final answer.

Accept equivalent forms, e.g. $\frac{1}{3} \ln(1 + \sin x) + \frac{2}{3} \ln(2 - \sin x) + C$, since $|\sin x - 2| = 2 - \sin x$.

A final answer left in terms of u earns the first six marks only.

FT: award the two final A1 marks from the candidate's own A and B , provided both are non-zero.

4. [Maximum mark: 7]

AA HL 2021 Nov P1 Q8

$$\frac{dy}{dx} + \frac{2y}{x} = \frac{\ln 2x}{x^2} \quad (\text{M1})$$

Attempt to find the integrating factor: (M1)

$$e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2 \quad (\text{A1})$$

$$x^2 \frac{dy}{dx} + 2xy = \ln 2x \Rightarrow \frac{d}{dx} (x^2 y) = \ln 2x$$

$$x^2 y = \int \ln 2x \, dx$$

Attempt to use integration by parts ($u = \ln 2x$, $dv = dx$) (M1)

$$x^2 y = x \ln 2x - x (+c) \quad \text{A1}$$

$$y = \frac{\ln 2x}{x} - \frac{1}{x} + \frac{c}{x^2}$$

Substitute $x = \frac{1}{2}$, $y = 4$ into an integrated equation involving c : M1

$$4 = 0 - 2 + 4c \Rightarrow c = \frac{3}{2}$$

$$y = \frac{\ln 2x}{x} - \frac{1}{x} + \frac{3}{2x^2} \quad \text{A1}$$

The M1 for substituting may be awarded before or after dividing by x^2 ; for $x^2 y = x \ln 2x - x + c$ it reads $\frac{1}{4}(4) = \frac{1}{2} \ln 1 - \frac{1}{2} + c$, so $c = \frac{3}{2}$.

Forgetting the constant c until after dividing, or not dividing c by x^2 (giving $y = \frac{\ln 2x}{x} - \frac{1}{x} + c$), loses the final A1.

Treating the equation as separable earns no marks.